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A scoping review of AI for sustainability and sustainable AI in higher education

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Abstract
Artificial intelligence (AI) is receiving increasing attention for its potential to support sustainability in higher education. As both users and developers of AI technologies, universities are well-positioned to advance environmental and social goals while ensuring that AI is used responsibly. This scoping review examines two related areas: AI for Sustainability, which involves using AI to achieve sustainability outcomes, and sustainable AI, which focuses on reducing the direct environmental and ethical impacts of AI itself. The review examines how these concepts are defined in literature, the field's evolution over time, and the potential or increasing application of AI in sustainability practices by universities. While there are promising examples of the use of AI to advance sustainability projects, including AI used for energy management, climate monitoring, and green campus programs, many of these efforts are still limited in scale and lack clear ethical or environmental guidelines. In addition, much of the current research is either conceptual or based on small-scale pilot projects, with few studies assessing long-term impact or full institutional adoption and commitment. The outcome of the review is intended to provide a structured overview of the extant of work in the area; identify research gaps, and introduce the concept of a future green campus. It argues that future research should focus on testing AI tools in real university settings, measuring their environmental impact, and developing policies that combine ethical and sustainability goals. Greater involvement from students, staff, and faculty will also be essential to ensure that AI supports a fair and sustainable future for higher education.

1 Introduction and background research

There are various conceptions of the meaning of sustainability. In this article, we refer to sustainability as meeting the needs of the present without compromising the ability of future generations to meet their own needs [113]. In a business context, sustainability involves integrating environmental, social, and economic responsibilities into an organization's operations and long-term strategies to ensure continued viability. This requires balancing profitability with social equity and environmental stewardship, ultimately generating positive outcomes [26, 28].

Research has also described business sustainability in terms of three main pillars. Environmental Sustainability focuses on protecting ecosystems, reducing greenhouse gas

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emissions, and conserving resources [40]. Social sustainability promotes fairness, inclusion, and community well-being [19]. Economic Sustainability emphasizes long-term profitability, ethical practices, and innovation [27]. Together, these dimensions ensure that organizations operate responsibly while addressing global issues such as climate change, inequality, and sustainable resource management.

Institutions of higher education are increasingly expected to embrace and address sustainability issues, for instance, through reducing their environmental footprint, promoting equity and inclusion, and managing limited resources responsibly [49]. These interconnected demands necessitate innovative, system-level approaches that surpass traditional methods, which are often too slow or limited to address global issues such as climate change, resource scarcity, and social inequality [53].

Artificial intelligence (AI) can offer potential support for addressing complex sustainability challenges that higher education institutions face. By processing large volumes of data, optimizing resource utilization, and enabling more inclusive and personalized learning environments, AI can enhance decision-making and operational efficiency across teaching, research, and administration [24, 118]. As AI becomes more deeply embedded within global institutional infrastructures, its importance to sustainability continues to grow. Others have argued that AI is a valuable tool capable of helping to accelerate and improve progress towards achieving the UN Sustainable Development Goals (SDGs). AI can foster innovation in various areas, such as the circular economy and smart city development, and acts as a vital link between sustainable development goals and effective strategies for climate change [50]. To capitalize on this potential, higher education institutions need to adopt AI to advance their sustainability agendas strategically. Strategic adoption involves the integration of AI technologies into the daily operations, decision-making, and activities of organizations and individuals within them [46]. This process encompasses the systematic selection, implementation, and utilization of AI systems to enhance efficiency, boost productivity, and strengthen competitive advantage [12]. The incorporation of AI into these sustainability frameworks represents not only technological innovation but also a strategic transformation in how universities pursue sustainable growth and operational excellence [4]. AI plays two complementary roles in supporting sustainability. The first, known as *AI for Sustainability*, focuses on using AI to address environmental and social challenges such as improving energy management and supporting climate adaptation [107]. The second, referred to as *Sustainable AI*, emphasizes developing and applying AI technologies in ways that are environmentally responsible and ethically sound [90].

Recognizing this dual relationship between AI and Sustainability is important because sustainability itself is a complex and multifaceted issue shaped by competing interests and evolving definitions [80]. Understanding sustainability in relation to AI, therefore, involves considering not only environmental concerns but also human relations issues such as ethics, fairness and the long-term success of AI applications, social and moral good that education is expected to foster [10].

The intersection of AI and Sustainability presents significant environmental challenges. One primary concern is the large amount of energy consumed during the training and operation of advanced AI models, which contributes to carbon emissions and environmental degradation [94]. For instance, training a single large language model such as GPT-3 can generate as much carbon dioxide as five cars over their entire

lifetimes, including fuel consumption [77]. Moreover, the increasing energy demands of data centres that power AI computations often rely on non-renewable energy sources, further amplifying their environmental footprint [87]. Although AI can support sustainability by enhancing energy systems and advancing climate change mitigation, its intensive energy and resource requirements underscore the importance of developing more sustainable AI practices.

Moreover, AI models require extensive hardware resources and high-performance computing infrastructure, which leads to increased electronic waste and material extraction for components such as GPUs and semiconductors. These processes further strain global supply chains and raise ethical concerns about sustainability and responsible technology use [13]. Consequently, while AI holds immense potential for driving innovation, addressing its environmental costs has become a pressing priority for researchers, policymakers, and technology developers seeking to promote sustainable AI practices.

Beyond environmental implications, the challenges surrounding AI also extend to broader ethical and social dimensions of sustainability, including fairness, transparency, and equitable access to technology. Concerns such as biased algorithmic outputs, the opaque nature of decision-making processes, and unequal access to AI tools raise critical questions about justice, accountability, and inclusion in AI-driven systems. These issues underline the need for the responsible design, deployment, and governance of AI within sustainability contexts [31, 43]. Nevertheless, growing research suggests that when developed and implemented responsibly, AI systems can act as powerful enablers of sustainability in higher education, supporting institutional goals related to environmental stewardship, social responsibility, and long-term resilience [67, 110].

1.1 Conceptual background

Since there is no shared definition of AI, in this article, we refer to AI as the development of machines, computer applications, and systems that can perform tasks requiring human intelligence, such as reasoning, learning, problem-solving, perception, and understanding language [61]. Modern AI systems exhibit varying degrees of autonomy, often learning and adapting over time through access to data [102]. These systems are typically built on four core abilities: perceiving, reasoning, acting, and learning. Together, these enable AI applications to sense their environment, make informed decisions, execute actions, and improve through experience (Fig. 1).

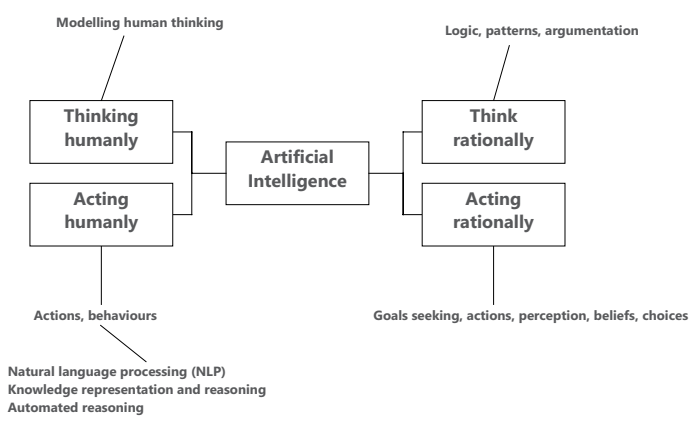

Fig. 1 Key dimensions of AI (adapted from [61])

The literature often highlights two primary dimensions of artificial intelligence. The first focuses on AI systems designed to exhibit human-like behaviour; machines that think and act as humans do [116]. This includes applications such as natural language processing (NLP) tools and virtual assistants (e.g., Siri, Alexa), chatbots, grammar checkers (e.g., Grammarly), translation services, and voice recognition systems. These technologies facilitate more natural human–machine interactions and are increasingly used to address sustainability challenges. For instance, AI can analyze satellite imagery for climate monitoring, track wildlife to prevent poaching, maximize agricultural resource use, and improve energy efficiency through smart grids and renewable energy systems [82, 110].

The second dimension emphasizes rational thinking and decision-making, focusing on developing systems capable of setting goals, reasoning logically, and improving performance over time [11]. This dimension is reflected in decision-support tools used in fields such as medicine, law, logistics, and finance, as well as in autonomous vehicles. These systems combine structured reasoning and data analysis to manage complex tasks, support predictive modelling, and enhance institutional and operational efficiency.

When applied with care, both human-like and rational dimensions of AI can contribute to outcomes that are efficient and ethically aligned. These capabilities enable the creation of systems that support long-term environmental responsibility, promote human values, protect human rights, and enhance institutional decision-making. As advancements in AI continue across various sectors, the opportunity to leverage AI for sustainable development and education becomes increasingly critical [98, 110].

1.2 The research knowledge gap and guiding review questions

Higher education institutions are uniquely positioned to shape the trajectory of AI and Sustainability. Universities are not only users of AI technologies but also key contributors to their development, evaluation, and ethical framing. As centres of research, innovation, and education, they can embed sustainability principles into AI curricula, develop solutions to environmental challenges, and prepare graduates to use AI in socially and environmentally responsible ways.

Research suggests that several universities have already begun adopting AI to support sustainability initiatives such as energy monitoring, waste reduction, and transport improvement, often in conjunction with smart infrastructure technologies [5, 7, 76]. However, these initiatives remain largely exploratory and context-specific, and their long-term impact on decarbonization, operational efficiency, and educational enhancement is underexplored.

An essential and emerging focus in Sustainability is Green AI for Green Campuses, which seeks to reduce the energy demands and environmental impact of AI systems [15, 20]. Green AI initiatives aim to improve algorithmic efficiency, develop low-power hardware, and enhance large-scale computing infrastructures [86]. These innovations have practical applications across various sustainability sectors, including smart grids, precision agriculture, and sustainable manufacturing [49].

Despite these advances, the Sustainability of AI itself remains a critical concern. Large-scale machine learning models require substantial computational resources and contribute significantly to carbon emissions [74, 107]. This paradox of AI being both a tool for sustainability and a potential source of environmental harm highlights the importance

of energy-efficient model design, lifecycle assessment, and responsible technological development [90, 96]. Ethical considerations further complicate the intersection of AI and Sustainability. As AI systems increasingly support decision-making in areas such as education, health, and governance, they risk increasing existing social biases and inequities. These challenges extend beyond technical limitations and reflect broader systemic issues embedded in the design and deployment of digital infrastructures [84, 90]. Research studies often make references to persistent algorithmic bias, data surveillance, violation of individual privacy, and unequal access to AI tools, which raise pressing concerns about fairness, justice, and inclusion [11, 48]. These ethical dilemmas challenge the values of equity and inclusivity central to the United Nations' Sustainable Development Goals.

Research has also mentioned that institutions adopting AI technologies face significant financial and operational barriers, especially when they require investments in hardware, software, technical expertise, and ongoing system maintenance [7, 9]. High initial costs, particularly for energy management and data infrastructure, are said to hinder widespread adoption, despite the potential long-term benefits [34].

Conceptually, the relationship between AI and Sustainability remains fragmented and inconsistently defined. However, the term sustainability has gained increasing coverage over the last decade but is often applied in varying contexts, reducing its conceptual clarity and practical utility [30]. Moreover, the relationship between the ethical and technical dimensions of AI, such as fairness, transparency, energy efficiency, and governance, has not been systematically explored within the context of the higher education sector. While isolated initiatives show promise, the extent to which AI contributes meaningfully to institutional sustainability goals, while enhancing teaching and research, remains unclear. In addition, developing sustainable AI requires balancing technological and societal benefits with environmental costs. Strategies such as promoting renewable energy use in AI operations, reducing data centre emissions, and investing in low-carbon, energy-efficient models are essential [35, 37, 87].

Despite increasing research and institutional interest, the intersection of AI and Sustainability within higher education remains insufficiently understood. Given the interdisciplinary and dynamic nature of this field, a scoping review is necessary to systematically map current evidence, identify knowledge gaps, and inform future research directions. Scoping reviews provide a rigorous methodological framework for examining emerging or complex topics by clarifying conceptual boundaries, synthesizing existing literature, and highlighting areas that warrant deeper empirical investigation [8, 55].

The primary objective of this review is to examine how AI and Sustainability intersect within higher education contexts. Specifically, the review seeks to establish a foundation for advancing research, informing institutional strategy, and supporting the formulation of evidence-based policy. Through a synthesis of existing studies, it aims to identify prevailing themes, emerging trends, and critical gaps that influence the ethical and sustainable integration of AI in higher education. Moreover, the review explores how AI can catalyze educational innovation, institutional sustainability, and broader societal benefits. Ultimately, this work aims to advance a coherent and comprehensive understanding of AI and Sustainability in higher education, providing actionable insights for researchers, policymakers, and institutional leaders.

To achieve these aims, the review adopts a systematic scoping approach guided by the methodological framework developed by Arksey and O'Malley [8] and further refined by Levac et al. [55] and the analysis, synthesis and presentation are guided by the systematic and tripartite approach (Table 1) (see [23]).

2 Methods and procedures

A generic literature review analyses existing studies to assess the depth and breadth of knowledge within a field [115]. It helps researchers avoid duplication [16], identify knowledge gaps, and justify the need for new research [85]. Literature reviews also support the development of research questions and conceptual frameworks [92, 100]. However, traditional approaches often lack methodological transparency, which limits comprehensiveness and introduces bias.

Systematic and scoping review methodologies offer structured, replicable approaches to address these limitations. A systematic review enhances rigour through transparent selection, evaluation, and synthesis, minimizing and ensuring reliability [57] (Gough et al. 2017). Researchers find systematic reviews valuable as they help decision-makers utilize the evidence obtained by systematically evaluating and appraising it [93]. Key features of a systematic literature review methodology are captured in Fig. 2.

Figure 2 presents a structured approach to conducting a systematic literature review, showing how research questions and prior literature inform the conceptual and procedural framing of the review. At the core is the development of explicit inclusion and exclusion criteria, which ensure transparency and guide the selection of relevant studies. The process then moves through clearly defined objectives, systematic selection, and two forms of analysis: description of key themes, synthesis and critique.

2.1 The scoping review method

A scoping review is a method within the systematic review methodology aimed at mapping the breadth and depth of a research topic [115]. It facilitates a synthesis of diverse perspectives from multiple literature sources, enabling a more comprehensive analysis of emerging trends and patterns, making it relevant for examining work on AI and Sustainability. Scoping reviews are particularly well-suited for exploring broad or emerging

Table 1 Guiding research questions and objectives

Research Question	Objective	Rationale
1. How are the terms *AI for Sustainability* and *sustainable AI* defined and conceptualized?	• To conceptualize and examine conceptual frameworks associated with the terms across disciplines	• Clarifying terminology is essential for establishing a coherent foundation, especially for interdisciplinary research, to reduce conceptual ambiguity
2. What is the evolution of research relating to AI and Sustainability?	• To trace the development of scholarly interest and identify key trends and thematic shifts over time	• Understanding the progression of research helps contextualize the current, underexplored areas that are highlighted and discussed
3. How are higher education institutions integrating sustainability initiatives guided by AI?	• To identify current practices, case studies, and institutional policies related to sustainable AI	• Examining real-world applications provides insights into how theory is translated into practice and informs institutional strategies
4. What are the key challenges, gaps, and opportunities identified in the literature for advancing research in AI and Sustainability in higher education?	• To synthesize reported and emerging opportunities and underdeveloped areas	• Identifying challenges and gaps informs future research directions and supports the development of targeted interventions

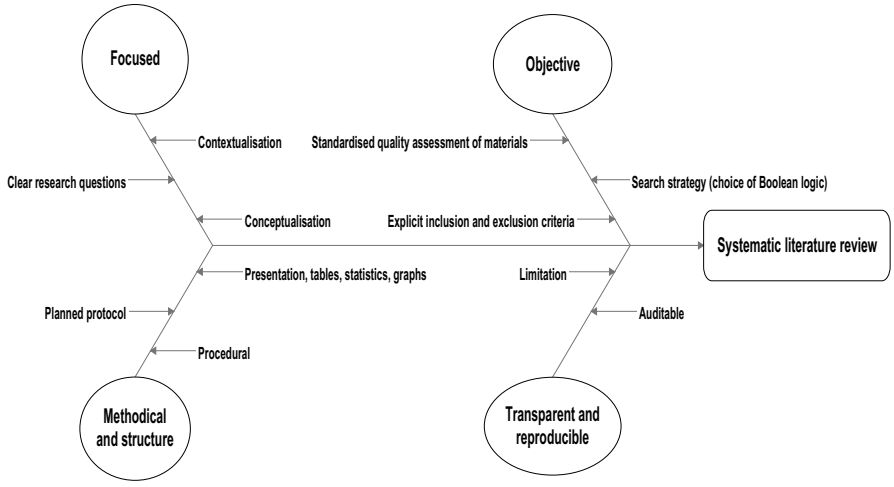

Fig. 2 Key features of a systematic literature review methodology

fields by mapping key concepts, identifying gaps, and clarifying definitions, without requiring a quality assessment. Unlike other systematic review methods, such as meta-analysis or meta-synthesis, that answer narrowly defined questions, scoping reviews address broader queries by identifying key concepts, types of evidence, and research gaps without critically appraising study quality [8, 78].

The current review supports conceptual clarification and the identification of underexplored intersections between AI and sustainability research in the context of higher education, thereby laying a foundation for future systematic reviews or empirical research [71, 104].

2.2 Procedures

The search process was iterative and involved multiple stages, including querying databases, retrieving results, assessing relevance, refining search terms, and selecting studies that aligned with the research questions. Given the interdisciplinary nature of the topic, which spans Computer Science, Environmental Studies, and Social Sciences, multiple academic databases were utilized to ensure broad coverage. These included Google Scholar, IEEE Xplore, ERIC for education, Web of Science for interdisciplinary research, Environment Complete, Business Source Complete and Scopus.

The following inclusion criteria were applied:

- Articles focused on Sustainability and AI, or Green AI in general, and higher education
- Empirical or theoretical studies published in a journal
- Studies with a well-defined methodology and scientific foundation
- Research articles written in English
- Publications dated between 1993 and 2025

Studies were excluded if they did not meet any of the following conditions:

- Not written in English
- Duplicates or extensions of already included studies
- Editorials, tutorials, books, extended abstracts, or similar formats
- Studies were also excluded if they lacked a clear, transparent methodology.

2.3 Search terms and Boolean strings

Search strings were developed based on keywords and the inclusion criteria. Boolean operators were used to broaden or narrow search results appropriately.

("Sustainable AI" OR "Green AI" OR "carbon-neutral computing" OR ("ethical AI" AND ("environment" OR "climate change" OR "sustainability")))
AND ("higher education" OR "university" OR "tertiary education")AND ("sustainability" OR "climate change" OR "environment" OR "ethics")
AND ("research" OR "case study" OR "empirical" OR "theoretical").

To ensure transparency, consistency, and methodological rigour, this review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-ScR) guidelines [106]. Given the interdisciplinary and evolving nature of the topic at the intersection of AI and Sustainability, adhering to PRISMA-ScR was essential for maintaining clarity and reproducibility in the scoping review process. Additionally, the core principles of the original PRISMA framework [66] were applied to enhance the standardization and quality of reporting. The PRISMA flowchart diagram illustrates the review process (see Fig. 3).

2.4 Data analysis and reporting

Data analysis and reporting followed the Systematic and Tripartite Approach (STA), illustrated in Fig. 4. The STA consists of three interconnected stages: description, synthesis, and critique [23]. This approach is particularly effective for analyzing complex and interdisciplinary research areas such as AI and Sustainability, as it provides a structured framework for organizing, integrating, and interpreting diverse sources of evidence.

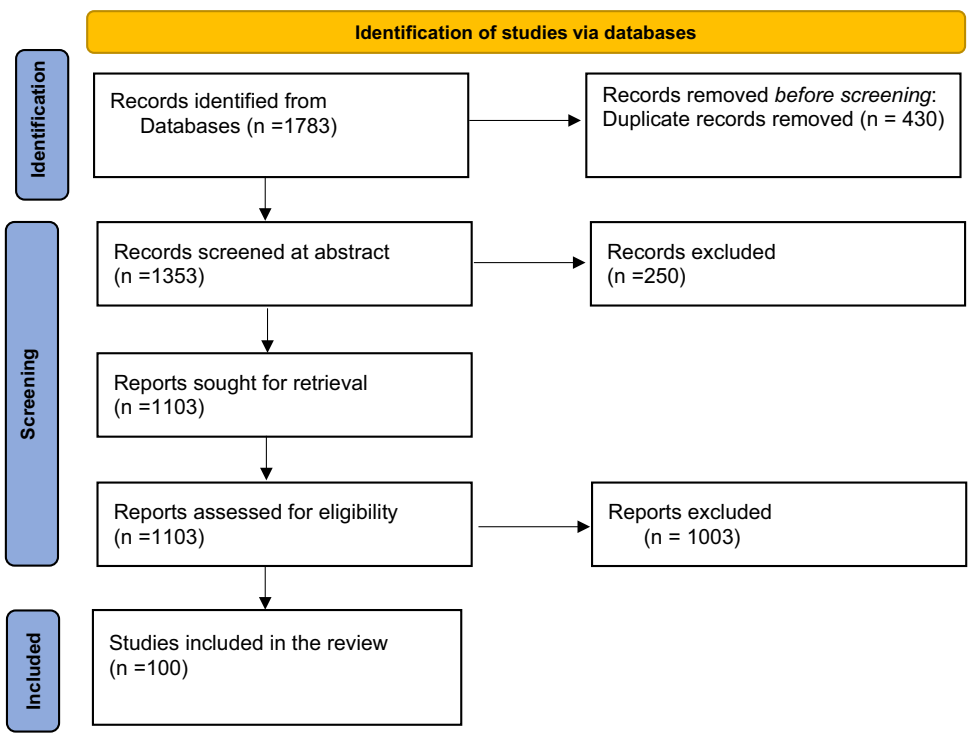

Fig. 3 PRISMA [75]

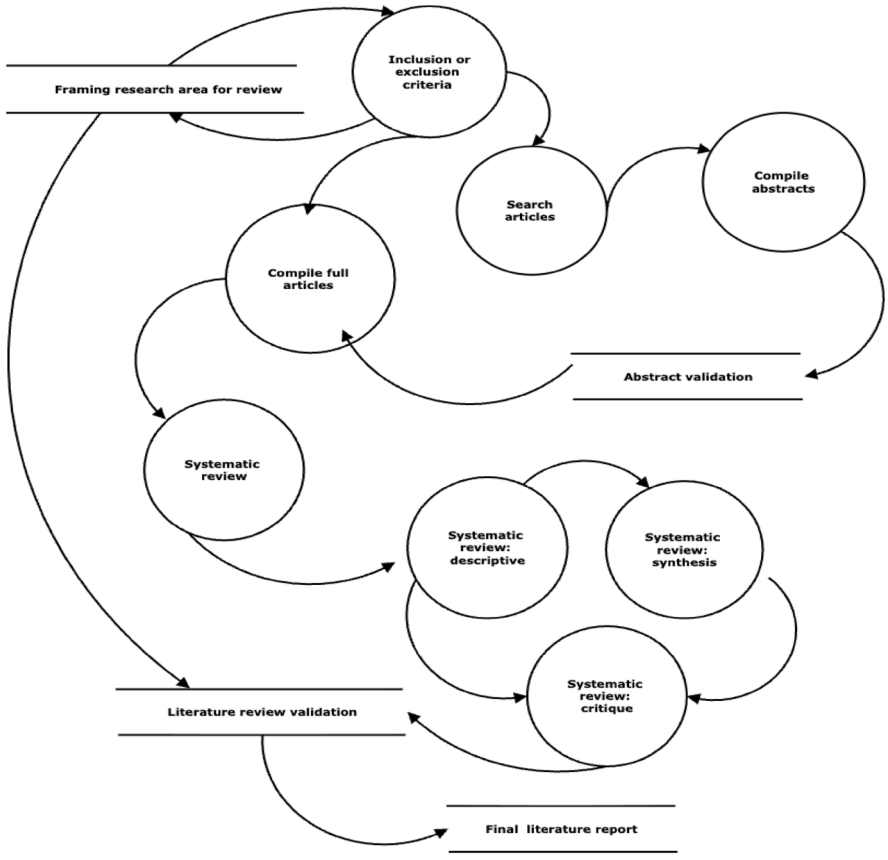

Fig. 4 Systematic and tripartite approach to literature review [23]

In the description stage, studies were mapped and categorized based on their focus, methodology, and outcomes. The synthesis stage involved identifying patterns, relationships, and emerging themes across the literature. Finally, the critique stage assessed the strengths, limitations, and gaps within the body of research, highlighting opportunities for future inquiry. By clearly delineating each phase, the STA ensures transparency and allows researchers to trace the analytical process from data interpretation to final reporting.

The STA methodology for conducting a literature review, as illustrated in Fig. 4, begins with *framing the review questions*. This initial step defines the scope and objectives of the review, setting the foundation for the subsequent stages of analysis.

The first stage, *description*, involves systematically presenting the findings from the collected literature, offering a detailed overview of current knowledge and highlighting research gaps in sustainable AI. The second stage, *synthesis*, integrates insights across studies to deepen understanding. This phase is central to constructing a coherent narrative around sustainable AI by identifying patterns, relationships, and areas of convergence or divergence in literature. The third stage, *critique*, evaluates the strengths, limitations, and underlying assumptions of the existing body of work. This stage also surfaces key themes and theoretical tensions, contributing to a more nuanced and critical interpretation of what constitutes sustainable AI.

3 Findings

Based on the PRISMA flow diagram, a total of 1783 records were initially identified through database searches. After removing 430 duplicate records, 1353 records remained for screening. Of these, 250 were excluded based on titles and abstracts, and 1103 were retrieved for full-text assessment. Following full-text review, 1003 were excluded. In total, only 100 studies met the inclusion criteria and were included in the final systematic review.

3.1 Overarching summary of key findings—descriptive

Findings from this scoping review reveals that research at the crossroads of artificial intelligence (AI) and sustainability is rapidly growing, focusing on both environmental and ethical issues. Two main themes are evident: AI for sustainability and sustainable AI. These themes show how AI can promote sustainability goals while also becoming more environmentally and ethically responsible. The review underscores the need to combine technological innovation, policymaking, and governance to ensure AI helps protect the environment and support social wellbeing. Examples like AI applications in sediment monitoring, disease tracking, and human rights surveillance illustrate its potential to aid ecological conservation and social justice [41, 59].

Within higher education, the review found that existing research mainly focuses on two areas. The first, AI for sustainability, investigates how AI technologies support environmental and social goals such as improving energy efficiency, enabling smart campus management, and assisting climate monitoring. The second, sustainable AI, considers how institutions can minimize the environmental and ethical impacts of AI systems by tackling issues like energy use, data governance, and algorithmic fairness.

Several initiatives have employed AI to support sustainability goals, but most remain limited in scope, exploratory, and not yet integrated into institutional strategies [119]. Much of the current work is theoretical, with limited empirical evidence on long-term results or widespread adoption across organisations [118].

The review also highlighted increasing institutional focus on Green AI, which emphasises energy efficiency, low-carbon computing, and responsible data use [31, 63]. This trend demonstrates a growing acknowledgement within higher education of the importance of aligning AI innovation with sustainability goals, ensuring that the development and implementation of AI systems are both ethically sound and environmentally sustainable.

3.2 Conceptions of AI and sustainability

The findings of this scoping review indicate that the role of AI in sustainability is framed through two complementary but distinct dimensions: AI for Sustainability and Sustainable AI. These represent the two dominant thematic orientations identified in the reviewed literature. AI for sustainability refers to the use of AI technologies to advance environmental, social, and economic goals. Studies in this area highlight the use of AI to improve energy efficiency, enhance climate forecasting, and support sustainable resource management [83, 110].

In contrast, Sustainable AI focuses on reducing the environmental and social impacts of AI itself. Research in this domain emphasizes the development of energy-efficient models, the need for fairness and transparency in algorithm design, and the importance

of promoting ethical and responsible use throughout the lifecycle of AI technologies [31, 107].

Although these two areas differ in focus, with one examining how AI can be applied to achieve sustainability goals and the other addressing how AI can be developed and used sustainably, they share a common emphasis on responsible innovation that aligns technological progress with long-term sustainability values. Table 2 summarises the key distinctions and intersections between these two dimensions, including their definitions, objectives, applications, stakeholders, and ethical considerations.

Overall, the findings suggest that research on AI and Sustainability in higher education remains conceptually rich but practically limited. Most studies focus on exploratory in nature or theoretical discussions rather than large-scale implementation or longitudinal evaluation. While universities are beginning to integrate AI to support sustainability goals, the evidence indicates a fragmented approach, with limited coordination between institutional strategies for digital innovation, environmental stewardship, and ethical governance. The dual focus on AI for Sustainability and Sustainable AI highlights both the opportunities and challenges facing higher education institutions as they seek to balance technological advancement with environmental and social responsibility.

3.3 Interdisciplinary trajectory and historical development of AI and Sustainability

The findings of this review highlight the interdisciplinary nature of AI and Sustainability, encompassing technological, environmental, and social dimensions. Early connections between AI and Sustainability were scattered, with few publications appearing in 1993, 2004, and 2008. From 2000 to 2014, the field entered a formative stage marked

Table 2 Comparison of AI for sustainability and sustainable AI

Aspect	AI for sustainability	Sustainable AI
Definition	• Use AI technologies to support and advance environmental, social, and economic sustainability goals	• Development and deployment of AI systems in environmentally, socially, and economically sustainable ways
Focus	• External impact: leveraging AI to solve sustainability challenges (e.g., climate change, resource management)	• Internal process: minimizing AI systems' negative impacts (e.g., carbon footprint, social harms)
Objectives	• Optimise energy use • Monitor biodiversity • Improve climate forecasting	• Reduce the energy consumption of AI models • Ensure ethical data use • Promote fairness and transparency
Applications	• AI-driven smart grids • AI for precision agriculture • AI in climate modelling	• Energy-efficient algorithm design • Low-carbon data centres • Fair and inclusive training datasets
Emphasis	• Application-level innovation	• Lifecycle and system-level sustainability
Stakeholders	• Environmental scientists, governments, NGOs, sustainability consultants, and leadership in higher education	• AI developers, tech companies, ethicists, policymakers
Challenges	• Access to quality data • Interdisciplinary collaboration • Bias in environmental models	• High energy consumption • Model opacity • Data privacy concerns
Ethical considerations	• Avoiding bias in sustainability metrics, Equitable access to benefits	• Transparency in design • Accountability for social and environmental impacts
Relation to SDGs	• Directly aligned with achieving Sustainable Development Goals (SDGs)	• Supports responsible innovation aligned with SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action)
Sustainability perspective	• AI as a tool for sustainability	• AI as an object to be made sustainable

by conceptual exploration and early theoretical framing. During this time, the concept of Green AI gained momentum, focusing on reducing the environmental costs of AI through energy-efficient algorithms, improved hardware design, and low-carbon computational workflows [3, 65, 88, 120]. Concerns about AI's energy use surfaced by 2010 and expanded by 2014 into broader discussions of AI's environmental footprint, leading to initial policy and institutional responses.

Between 2019 and 2020, the discussion evolved through the introduction of sustainability-by-design frameworks [88], and the notion of Sustainable AI became more prominent—recognizing AI's dual role as both a potential contributor to and mitigator of sustainability challenges [18]. Applications of this perspective now include intelligent energy grids, precision agriculture, low-emission logistics, and smart infrastructure [33, 36, 39]. Research has also expanded to address responsible data governance and AI alignment with the United Nations Sustainable Development Goals [82], emphasizing renewable energy integration, reduced ecological impact, and embedding sustainability across the AI lifecycle.

Figure 5 provides a visual summary of this progression, illustrating the growing maturity and diversification of AI and sustainability research since the early 2000s.

3.4 Evolution of Green AI and the expanding landscape of sustainable AI research

The progression of research at the intersection of AI and environmental issues, focusing on the emergence and evolution of Green AI. The timeline begins with the foundational phase (2000–2010), which laid the theoretical groundwork for linking AI technologies with environmental concerns. This was followed by a period (2010–2014) marked by a significant increase in understanding and recognition of the energy demands of AI systems, as well as the accompanying ethical implications, including opacity, bias, and unregulated use.

Between 2014 and 2019, concerns about the ethical and environmental impact of AI, particularly regarding privacy, fairness, and energy use, gained broader recognition within academia, industry, and civil society. This led to a wave of institutional responses during 2019–2020, including policy proposals, industry-led accountability initiatives, and calls for transparency in model training costs and the use of AI in critical domains such as finance, health, and education.

From 2020 to 2025, research has increasingly focused on the practical applications of sustainable AI and AI for Sustainability, emphasizing the development of energy-efficient

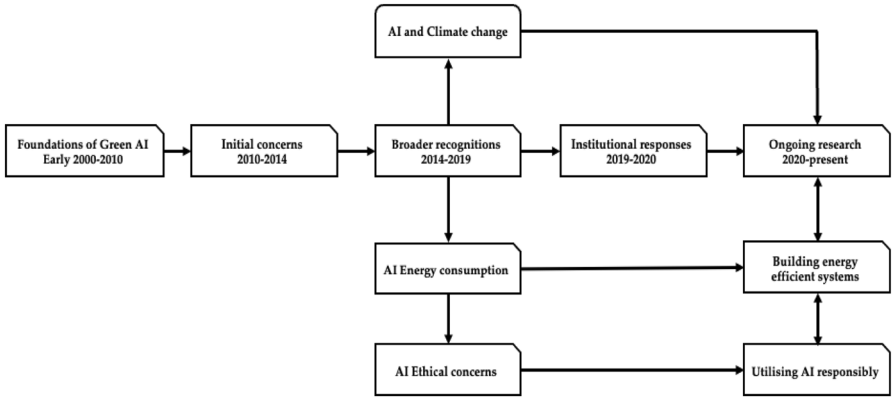

Fig. 5 Evolution of research discourses on AI and sustainability

architectures, optimization of algorithms, and advancement of responsible AI practices. This phase marks a transition from conceptual exploration to implementation-focused research, aiming to mitigate AI's ecological footprint and align it with sustainability goals.

3.5 Disciplinary and cross-sectoral contributions to sustainable AI

As summarised in Table 1, Sustainable AI research spans a diverse range of disciplinary and cross-sectoral domains. The core technical disciplines are computer science, environmental science, electrical engineering, and data science, forming the foundation of this research landscape. These fields contribute to designing energy-efficient algorithms, developing sustainable hardware systems, and implementing responsible data management practices [94].

In addition to these core areas, interdisciplinary domains such as AI ethics and policy, sustainable business models, and educational technology play a crucial and indispensable role in framing the interface of Sustainability and AI within broader societal and institutional contexts. These domains promote the incorporation of ethical frameworks, regulatory considerations, and curricular innovation to ensure the responsible and equitable adoption of AI technologies [43]. Emerging trends are also shaping the future direction of sustainability and the role of AI. Innovations such as Green AI, AI in renewable energy systems, and AI applications for environmental protection reflect a growing emphasis on using AI not only more efficiently, but also more purposefully to address climate and ecological challenges [82, 101, 110].

Furthermore, strategic collaborations among academia, industry, governments, and international organizations are increasingly recognized as essential for scaling and operationalizing Sustainable AI. These partnerships enable global knowledge exchange, regulatory alignment, and coordinated efforts to develop and deploy sustainable AI technologies [17, 110, 112].

As shown in Table 3, this scoping review reveals that research on AI for sustainability and sustainable AI spans multiple disciplines. Fundamental fields such as Computer Science, Environmental Science, Electrical Engineering, and Data Science form the technical basis, emphasising algorithm efficiency, hardware optimization, and data governance [94]. Interdisciplinary domains like AI Ethics, Sustainability Policy, and Educational Technology bridge technical innovation with ethical and policy concerns [43]. Emerging trends, including Green AI, AI in Renewable Energy, and AI for Environmental Protection, highlight increasing efforts to minimise AI's environmental impact and harness

Table 3 Disciplines that have contributed to AI for sustainability and sustainable AI research

Category	Areas of focus	Description
Core disciplines	Computer Science, Environmental Science, Electrical Engineering, Data Science	• Focuses on the technical foundations of sustainable AI, including algorithm design, hardware optimization, and data governance [94]
Interdisciplinary areas	AI Ethics and Policy, Sustainable Business Models, Educational Technology	• Explore the integration of ethical principles, sustainability policy and business models, and educational frameworks [43]
Emerging trends	Green AI, AI in Renewable Energy, AI for Environmental Protection	• Highlights innovative efforts to reduce AI's environmental footprint and apply AI to sustainability goals [82, 101, 110]
Strategic collaborations	Academia–Industry Partnerships, Government Regulation, International Cooperation	• Supports the global scaling of Sustainable AI through cross-sector partnerships and policy alignment [17, 110, 112]

AI for sustainability objectives [82, 101, 110]. Collaboration among Academia, Industry, and Government promotes global cooperation, regulation, and scaling of sustainable AI initiatives [17, 110, 112].

3.6 Trends in research output and the maturity of the field

As shown in Fig. 6, research activity on AI and Sustainability saw a notable growth beginning in 2015. This growth can be attributed to several global and technological milestones, including the signing of the Paris Agreement, a significant event that raised international attention to sustainability and climate action. The peak in publication output observed in 2021 reflects the rapid advancement of AI technologies, especially in deep learning, and the consolidation of Sustainable AI as a legitimate and urgent research agenda.

Figure 6 displays the increase in publications on AI and sustainability from 1995 to 2025. There was little activity until 2015, after which research output grew rapidly. This trend indicates a growing awareness of AI's ability to help solve environmental and social issues, as well as ongoing concerns about its ethical and environmental impacts.

Understanding this publication trajectory is crucial because it shows how AI and Sustainability have developed as an interdisciplinary field. It highlights how global concerns about climate change and the need for responsible innovation have influenced academic research and the role of higher education in promoting sustainable AI practices.

As shown in Fig. 7, there are three major themes in AI and sustainability research from 2015 to 2025: Ethical AI, Responsible AI, and Energy-efficient AI. Ethical AI gained traction around 2015 as researchers began to examine issues of fairness, transparency, and algorithmic bias in automated decision-making [43]. By 2020, Ethical AI had reached a 50 per cent trend presence, maintaining that level through 2025, indicating its central role in shaping the ethical development and governance of AI systems.

Responsible AI emerged around 2018, focusing on translating ethical principles into practical actions and policies. It emphasizes accountability, inclusivity, and transparency in the deployment of AI systems. By 2020, Responsible AI had reached parity with Ethical AI in trend presence, reflecting the growing focus on aligning AI development with social and environmental priorities [110, 112].

Energy-efficient AI appeared around 2020 as attention turned to the environmental costs of large-scale machine learning. This trend highlights efforts to reduce AI's carbon footprint through greener algorithms, improved computational efficiency, and the

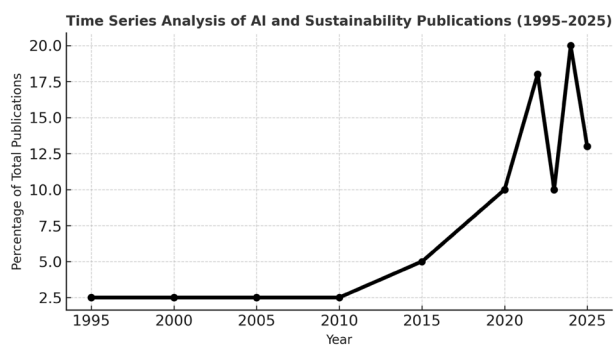

Fig. 6 Publications trends on AI and sustainability (1993–2025)

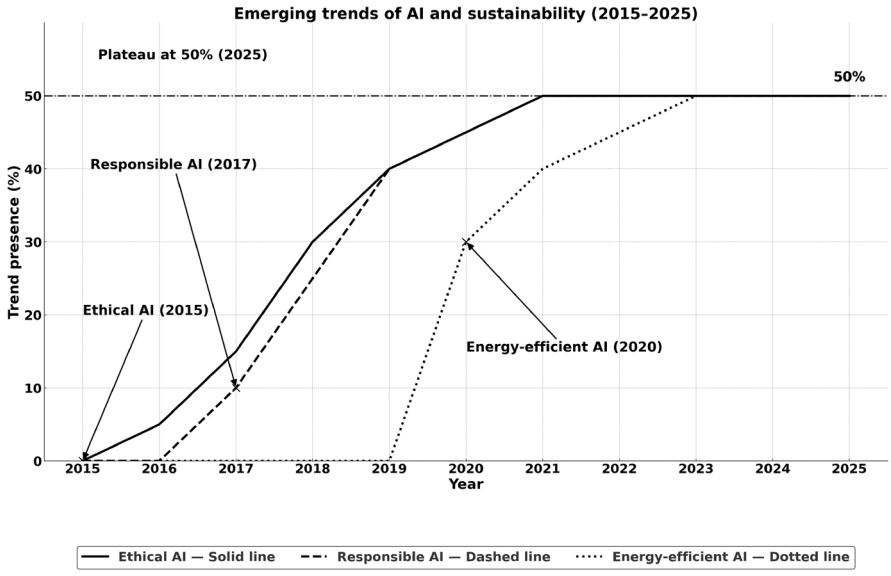

Fig. 7 Emerging AI and sustainability themes (2015–2025)

integration of environmental performance metrics [82, 94]. By 2022, this trend had also reached a 50 per cent presence and remained stable through 2025.

Collectively, these trends reflect a maturing research landscape that moves beyond technical advancement to address broader ethical, social, and environmental dimensions. Their convergence around 50 per cent by 2025 suggests that the discourse on AI has become more integrated, emphasizing the shared goal of developing responsible, fair, and sustainable AI systems that respond to global sustainability challenges.

4 Synthesis

4.1 AI for sustainability

The review identified consistent evidence indicating that the training and maintenance of large-scale artificial intelligence models consume considerable energy, resulting in noteworthy carbon emissions. Research reports that the carbon footprint associated with training a single deep learning model may be comparable to that of several automobiles over their entire lifespan [94, 107]. Such emissions are predominantly generated by the electricity requirements of data centres and specialized AI hardware, which often depend on non-renewable energy sources [44]. Additional energy expenditure is incurred in regulating temperature and preserving the computational infrastructure [60, 62].

A primary focus within the reviewed literature is the development of energy-efficient and low-carbon AI systems. Researchers recommend algorithmic improvements, renewable energy-powered infrastructure, and policy mechanisms that promote sustainable AI design [47, 60]. Technical innovations such as deep compression [38] and EfficientNet [99] have been demonstrated to diminish computational demands, while new AI chip designs aim to reduce power consumption [101]. The Green AI movement, which incorporates environmental considerations into AI research and development, has gained significant prominence for advocating carbon-conscious algorithms and responsible energy utilization [2, 35, 37, 87, 112, 121].

The current review identified increasing deployment of AI technologies in supporting sustainability initiatives in general, including environmental monitoring, climate modelling, and resource optimization. Examples encompass AI systems utilized for tracking deforestation, wildlife conservation, and illegal fishing activities [45]; predictive tools that improve water and energy utilization in agriculture and urban planning [91],and models designed to forecast flooding events and sea-level variations [74]. In the domain of higher education, artificial intelligence is being incorporated into campus operations through intelligent energy management systems and predictive maintenance, thereby resulting in reduced emissions and enhanced operational efficiency [117].

Another significant finding is the increasing body of research exploring the role of AI in advancing the United Nations Sustainable Development Goals (SDGs). Research indicates that artificial intelligence positively contributes to 128 of the 169 SDG targets, particularly in areas such as health, education, environmental monitoring, and urban development [50–52, 58, 110].

These contributions are particularly evident in sectors such as healthcare, where AI supports disease detection and public health management [29, 105], and education, where AI enhances personalized learning and improves accessibility [42, 118]. Collectively, these studies demonstrate AI's potential as a transformative enabler of sustainability, promoting technological innovation and societal well-being when applied responsibly and ethically.

However, the review reveals uneven attention across the Sustainable Development Goals (SDGs), with significant focus on SDG 9 (industry, innovation, and infrastructure) but limited engagement with SDG 4 (quality education) and SDG 14 (life below water) [72, 73]. Persistent barriers, including unequal access to technology and disparities in digital infrastructure, continue to hinder the equitable and widespread adoption of AI for sustainable development.

Overall, the findings of this scoping review show an emerging overlap between AI for Sustainability and Green or Sustainable AI, both aiming to align technological progress with environmental responsibility and economic efficiency. While AI for Sustainability focuses on using AI to support sustainable goals such as energy management, climate action, and resource optimization, Green AI emphasizes reducing the environmental impact of AI technologies themselves by improving energy efficiency and computational sustainability [82, 87].

This overlap suggests a two-way relationship in which AI contributes to sustainability while also requiring more sustainable development practices. The integration of these perspectives highlights growing attention to how AI can both advance and embody sustainability principles. The review also points to the need for stronger governance frameworks that ensure responsible, transparent, and ethical AI practices aligned with global sustainability objectives [72, 110].

4.2 Sustainable AI

The findings from this scoping review reveal a growing integration of ethical, social, and environmental considerations in research on AI. Across the literature, there is an increasing recognition that AI should be developed and utilized in ways that align with ethical principles and sustainability goals. Common concerns include data privacy, algorithmic bias, surveillance, and limited transparency in automated decision-making.

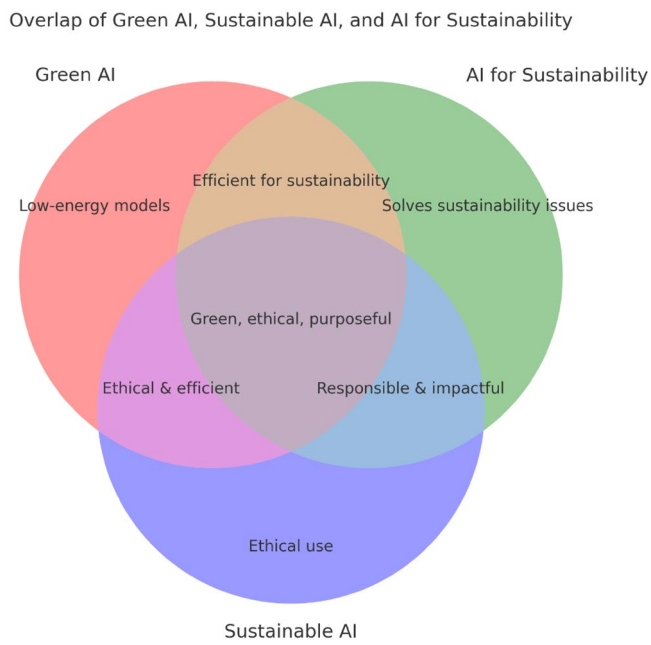

Fig. 8 The overlap of the key areas explored in this current review

These issues are especially critical in sectors such as healthcare, finance, and criminal justice, where AI systems impact vital human outcomes [43, 64, 79].

The review further indicates that discussions on ethical and responsible AI are becoming central to debates about sustainable technology. Ethical AI emphasizes fairness, accountability, and respect for human rights, while responsible AI puts these principles into practice through governance frameworks and policy mechanisms [25, 70]. International initiatives, including those led by the European Union and the OECD, have helped shape these governance models and promote greater institutional accountability in AI development and deployment.

Building on these perspectives, the review emphasizes the importance of governance frameworks that clearly incorporate ethical standards alongside sustainable development goals. These frameworks can assist in balancing AI's environmental impacts, such as high energy consumption and resource use, with its social and economic advantages [14, 97, 108].

The scoping review shows that Green AI, Sustainable AI, and AI for Sustainability are interconnected concepts (Fig. 8) that together define the emerging relationship between artificial intelligence and sustainable development. The intersection between Green AI and Sustainable AI reflects efforts to reduce AI's environmental footprint while promoting fairness, transparency, and accountability in its design and use. Where Green AI overlaps with AI for Sustainability, the emphasis is on applying energy-efficient AI systems to advance sustainability goals, such as improving energy management, supporting climate action, and optimizing resource use.

The convergence between Sustainable AI and AI for Sustainability highlights the use of socially responsible and ethically grounded AI to address complex environmental and societal challenges. At the centre of these relationships is a shared vision of AI that is efficient, ethical, and sustainability oriented. This integrated framework provides a

meaningful basis for guiding responsible innovation, shaping institutional strategies, and fostering sustainable AI practices in higher education and beyond.

From a critical perspective, although the existing literature portrays AI as a promising instrument for sustainable development, it frequently underrepresents the systemic and contextual obstacles associated with its deployment. The prevalent optimism regarding AI's role in higher education and the promotion of sustainability neglects considerations of human and material costs, ethical issues, and the unequal distribution of benefits across different regions and sectors [107]. Furthermore, the dominance of research originating from the Global North, the limited incorporation of indigenous and community knowledge, and the lack of long-term impact data expose significant gaps that warrant future attention.

A more balanced approach that critically examines both AI's potential for digital transformation in higher education and its socio-environmental risks is essential to avoid inequalities in sustainability efforts. This approach should promote inclusive, equitable, and environmentally conscious innovation [25].

5 Challenges and opportunities

This scoping review concludes that while AI possesses considerable potential to advance sustainability goals within higher education, its comprehensive integration into existing institutional frameworks remains somewhat constrained. Technologies such as machine learning have demonstrated potential in optimizing energy consumption and managing campus resources [69]. Nevertheless, the broader application of AI for sustainability has yet to be fully realized due to existing barriers, including limited funding, underdeveloped infrastructure, and unresolved ethical concerns, particularly about algorithmic bias, data governance, and transparency [6, 68, 89].

The literature confirms that AI can contribute significantly to the SDGs by supporting inclusive, ethical, and environmentally responsible practices [50–52, 109]. However, most current applications remain at the conceptual or pilot stage. Achieving AI's transformative potential in higher education will require sustained investment, policy alignment, and interdisciplinary collaboration.

The intersection of AI and sustainability encompasses energy-efficient computing, renewable-powered data centres, and sustainable hardware design [87, 94]. A recurring theme within this review pertains to the environmental impacts associated with AI systems. Studies underscore the substantial energy consumption required for training large-scale models, thereby raising concerns regarding carbon emissions attributable to AI and their implications for long-term environmental sustainability [87, 94]. Luccioni et al. [56] highlight that general-purpose generative AI systems consume significantly more resources compared to task-specific models, further exacerbating their environmental impact.

Furze [32] emphasized the resource requirements during both the training and inference phases of developing and deploying generative models, such as GPT-4, and noted the absence of educational strategies for their responsible integration. Drawing on work such as Crawford [22], Furze conceptualizes AI as an extractive technology that significantly impacts environmental systems. More broadly, there is increasing evidence within the literature supporting the incorporation of sustainability principles into AI governance frameworks and educational policies [31, 110].

Against this backdrop, the review has identified Sustainable AI as a valuable framework for addressing the intersecting ethical, environmental, and economic dimensions of AI development. The literature emphasizes the responsible use of AI to promote social good, environmental stewardship, and inclusive economic growth, all of which are critical aspects of sustainable AI.

Ethical concerns concentrate on reducing bias, promoting transparency, and protecting data privacy [31]. From an economic viewpoint, priorities include fair access, supporting employment through human-in-the-loop systems, and the long-term sustainability of AI infrastructure. Research also highlights the potential of AI in environmental monitoring, resource management, and emissions reduction [45, 91], although institutional strategies to incorporate these tools into sustainability planning are still developing.

Despite increased focus on Sustainable AI, governance frameworks in higher education often overlook environmental aspects, raising the risk of unintended ecological effects [97]. Furthermore, there is limited research on the medium and long-term environmental impacts of AI systems, and issues such as transparency and accountability remain insufficiently examined in the higher education sector [54].

The review also emphasizes ongoing global disparities in AI preparedness. Technologically advanced nations are more suitably positioned to capitalize on AI innovations, whereas developing countries frequently lack the essential infrastructure and capacity, thereby exacerbating existing digital divides [18]. Addressing these disparities will necessitate targeted assistance for under-resourced institutions, encompassing inclusive education, open-source tools, and comprehensive data protection frameworks.

Although AI has shown potential to improve accessibility, operational efficiency, and sustainability assessment in higher education [1], its role in advancing broader sustainability initiatives remains underexamined. A significant gap remains in understanding how AI ethics, environmental sustainability, and institutional governance intersect in higher education [21, 31, 50–52]. Without an integrated approach, institutions may overlook key trade-offs between energy efficiency and data privacy, or between social benefits of AI and environmental cost [64, 94].

Addressing ethics, sustainability, and governance as isolated issues may result in fragmented and ineffective responses to the escalating complexity of artificial intelligence [109]. Therefore, this review endorses integrated strategies that synthesize ethical considerations, environmental concerns, and data governance perspectives to support the development of cohesive institutional policies for the responsible and adaptive deployment of AI.

Findings also highlighted the importance of robust governance mechanisms, including regulatory standards, international policy coordination, and meaningful public engagement in AI oversight [25, 97]. Scholars also endorse integrating ethical assessment throughout the various stages of the AI development lifecycle, from design to deployment, to ensure fairness, transparency, and accountability [25, 31, 70]. However, there remains a lack of concrete frameworks or tools to assess institutional readiness for AI-driven digital transformation in higher education.

6 Conclusion, limitations and future research

This scoping review examines the intersection of artificial intelligence and sustainability within higher education by investigating two interconnected yet distinct pathways: sustainable artificial intelligence and artificial intelligence for sustainability. The findings provide insights into the definitions of these concepts, the evolution of research in this domain, the engagement of higher education institutions with AI to promote sustainability, and the remaining gaps for future research.

We recognize that, given the interdisciplinary scope of this topic, it is vital to provide a clear and transparent rationale for database selection. While Google Scholar, IEEE Xplore, and Scopus are valuable resources that offer strengths in technical fields, broad multidisciplinary indexing, and extensive coverage, they are insufficient on their own to encompass all relevant research. Acknowledging these limitations, especially for a field spanning computer science, environmental studies, education, and policy, we therefore expanded the search to include ERIC for education, Web of Science for interdisciplinary research, and specialized databases such as Environment Complete and Business Source Complete.

Another significant limitation of this study is that most research on using AI for Sustainability in higher education comes from the Global North, mainly North America and Europe, with little input from the Global South. Apart from a few studies in South Africa, evidence from other regions remains scarce. Future research should explore how institutions in the Global South employ and adapt AI to promote sustainability within their unique social, economic, and environmental settings.

Lastly, while systematic reviews aim to answer a specific question through critical appraisal and synthesis of high-quality evidence, scoping reviews are more extensive in scope. They seek to map existing literature, clarify key concepts, and identify research gaps, without assessing study quality. Given the emerging and conceptually diverse nature of Sustainability and AI initiatives in higher education, this review aims to outline critical areas that need further development and exploration. This review is intentionally broad to emphasize key areas for future research and investigation.

Q1. How are the terms' AI for sustainability' and' sustainable AI' defined and conceptualized?

The literature characterizes AI for sustainability as the utilization of artificial intelligence technologies to promote environmental, social, and economic sustainability objectives. This encompasses applications such as climate modelling, biodiversity monitoring, precision agriculture, energy management, and sustainable urban planning [45, 91]. AI is regarded as an enabling instrument to facilitate the United Nations Sustainable Development Goals by enhancing efficiency, improving decision-making processes, and supporting extensive data-driven environmental surveillance.

Sustainable AI, on the other hand, involves efforts to minimize the unintended and negative environmental, ethical, and social impacts linked to the development and deployment of AI systems. This includes reducing energy consumption and carbon emissions associated with training and deployment, as well as tackling issues such as algorithmic bias, lack of transparency, and accountability [62, 79, 95, 107, 114] (Falks et al. 2024). Sustainable AI encourages the development of energy-efficient algorithms, the utilization of renewable energy in data centres, and the implementation of ethical governance mechanisms.

Although these concepts aim to align AI with sustainability goals, they are often tackled separately rather than through integrated approaches. This separation hampers the development of comprehensive theoretical and operational frameworks. Bridging these concepts is crucial to addressing the interdependence between designing sustainable AI systems and applying AI to sustainability challenges.

Q2. How has research on AI and Sustainability evolved?

The review shows that scholarly interest in AI and Sustainability has steadily grown since 2015. Early research mainly focused on ethical issues and potential risks, while recent studies tend to emphasize energy efficiency and the environmental impacts of AI. At the same time, practical applications of AI for Sustainability, such as in agriculture, climate change mitigation, and resource management, have become more visible. However, many of these initiatives remain in early phases. There is a significant need for research that critically examines the broader social, ethical, and ecological implications of AI technology beyond just the technical aspects.

Q3. How are higher education institutions incorporating sustainability initiatives through AI?

AI integration in higher education remains in its early stages. Some institutions are experimenting with AI in green campus projects, digital learning environments, and resource management systems. However, these efforts are often disconnected and do not align with the institutions' sustainability goals. Most universities have yet to establish comprehensive governance frameworks that embed ethical and environmental considerations into their AI strategies. As both creators and users of AI technologies, higher education institutions are uniquely positioned to demonstrate responsible innovation, but they must progress from sporadic implementation to systemic, value-driven approaches.

Q4. What are the main challenges, gaps, and opportunities identified in the literature for advancing research in AI and Sustainability in higher education?

Key challenges include undeveloped infrastructure, limited regulatory guidance, and a lack of integrated ethical and environmental standards. Empirical studies assessing the long-term environmental and social impacts of AI in educational settings are scarce. Additionally, global disparities in AI readiness are likely to worsen existing inequalities [18]. The literature highlights significant opportunities, including the creation of low-carbon AI systems, stakeholder-led design approaches, and the positioning of universities as testbeds for ethical and sustainable AI practices.

6.1 Outstanding issues and future research

As universities increasingly prioritize sustainability, the green campus initiative presents a valuable opportunity to explore how artificial intelligence can support key objectives, including energy optimization, waste reduction, and online education. This scoping review highlights both the potential benefits and the challenges involved in promoting sustainability through artificial intelligence within higher education. On one hand, AI presents distinct opportunities to advance sustainability goals, such as improving energy efficiency, improving resource management, and optimizing operational practices. However, its integration into institutional systems remains limited, with most applications at the pilot stage and few strategies in place for long-term implementation.

Meanwhile, sustainable AI highlights the ethical and environmental effects of AI itself. High energy use, lack of transparency, and disjointed governance structures pose risks

that could hinder sustainability efforts. To tackle these issues, organizations need to adopt a holistic approach that combines ethical, environmental, and data governance viewpoints. Only then can AI serve as both a tool for sustainability and a form of sustainable technology.

Based on the current scoping review, several key research gaps have been identified that will require further investigation:

- Empirical studies validate Sustainable AI models in real-world higher education contexts, including assessments of energy use and resource impact.
- Developing integrated governance frameworks that combine ethical and environmental considerations in institutional AI implementation.
- Examine the long-term environmental impacts of AI technologies, particularly within academic and research-intensive settings.
- Assessment of the extent to which higher education institutions are equipped to fully embrace the transformative potential of AI in addressing sustainability challenges.
- Strategies for equitable and inclusive AI adoption that address global disparities in access, capacity, and infrastructure.
- Cross-disciplinary research exploring the interaction between ethical, environmental, and operational goals in AI-driven sustainability efforts.
- Integrating AI into sustainability education, including curriculum development, pedagogical approaches, and student engagement.
- Policy and regulatory analysis to ensure AI development aligns with the Sustainable Development Goals (SDGs) and institutional green campus strategies.
- Design of stakeholder engagement frameworks that actively involve faculty, students, and administrators in AI-related decision-making.
- Addressing these research areas is critical for advancing the responsible integration of AI in higher education and ensuring that AI contributes meaningfully to long-term sustainability outcomes.

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BD designed the review, led the methodology, literature search, analysis, and synthesis of findings, and drafted the original manuscript. NP and SC supervised the study, provided guidance on the review framework and structure, and contributed to the critical revision and final editing of the manuscript. All authors equally contributed to the final article.

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Clinical trial number
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Declarations

Ethics approval and consent to participate
This is a scoping review of publicly available published studies and does not involve direct interaction with human subjects, animal subjects, or the collection of private or sensitive data information. The review adheres to the ethical principles outlined by the authors' institution for secondary research. The study aligns with the University of Otago's ethical research standards. All review data were drawn from academic databases, journal articles, conference proceedings, and publicly available research repositories. Data collection included only information reported in the original sources. No personal or sensitive data will be gathered or created. Not applicable. This project consists of a systematic scoping review of published literature and publicly available sources. No human participants are involved, and no personal or identifiable information is collected. Therefore, informed consent is not necessary.

Consent for publication
This manuscript relies solely on previously published and publicly accessible research. No human participants were involved, and no personally identifiable data was collected. Consequently, consent to publish from individual participants was not necessary. All sources are properly cited following scholarly standards. All authors have consented to publish.

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